WEEK 3 · DAY 1 · OVERVIEW

Treasury Curve Interpolation



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ABOUT 40 MIN WITH EXERCISES / 7 SECTIONS / 4 EXERCISES

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The Desk Lead's Note

Week 3 measures every bond against Treasuries, so each bond needs a Treasury yield at its own maturity. Treasury publishes fourteen tenors. Read me the 30 September curve at 6.4 years, then put the method in bondmath and tie it to Excel. Tell me which curve our holdings were built on, and how much a straight line between tenors moves the number.

ABOUT 40 MIN WITH EXERCISES / 7 SECTIONS / 4 EXERCISES

Your Route Through Day 1

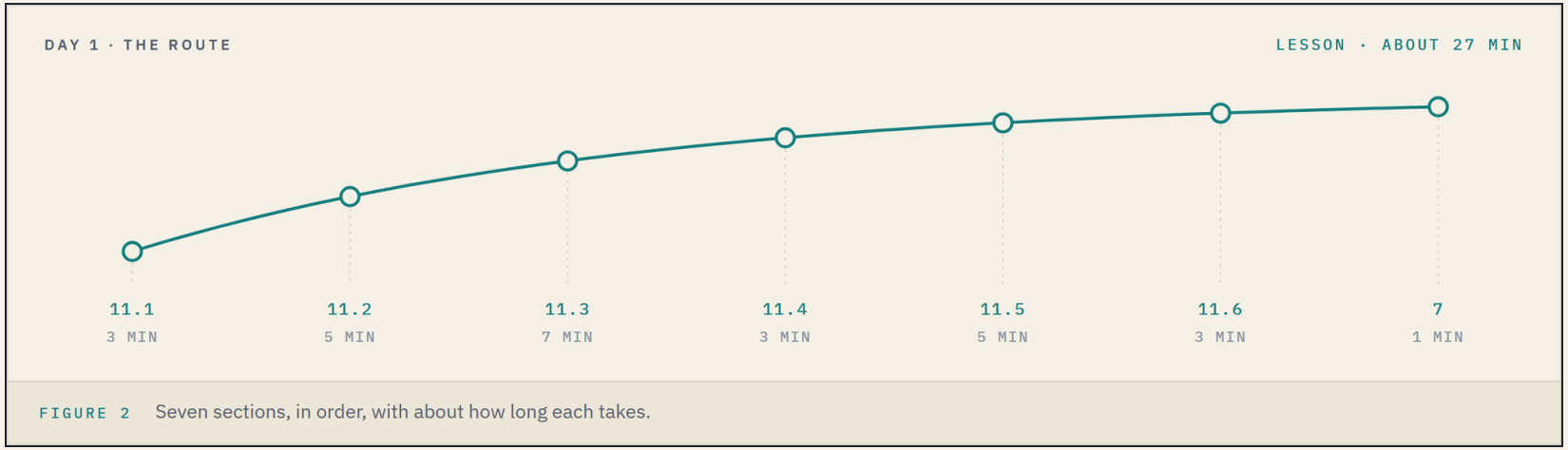


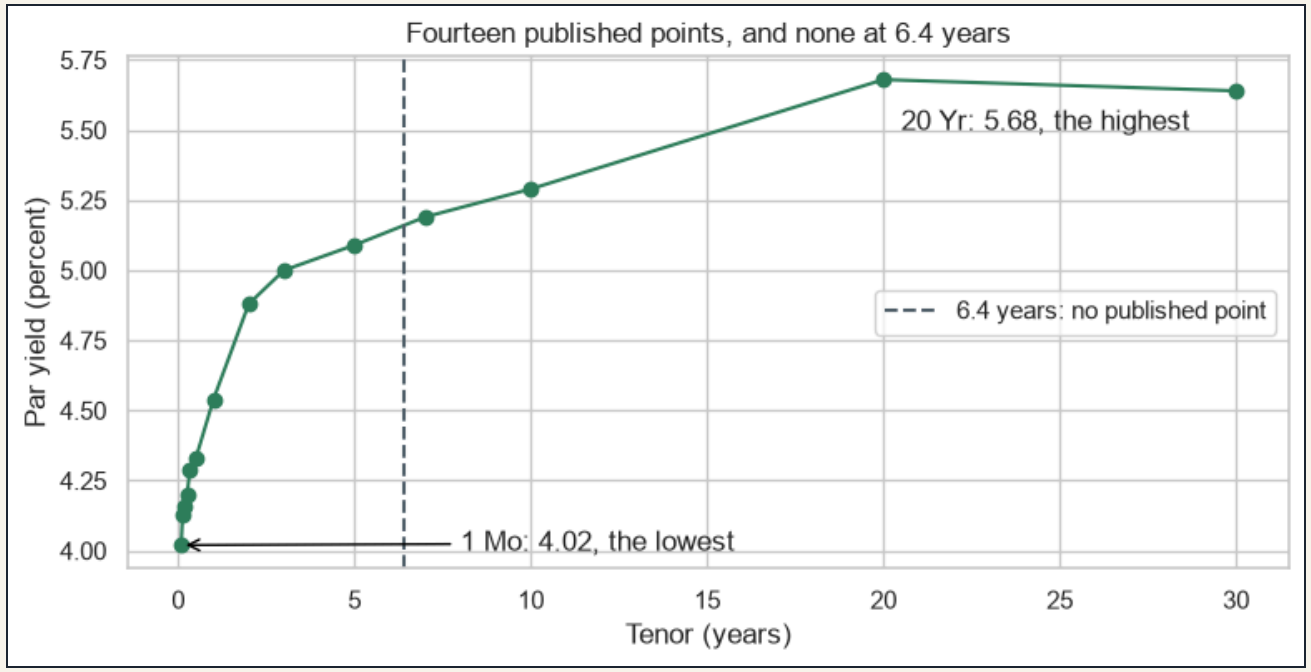
FIGURE 2 Seven sections, in order, with about how long each takes.

	SECTION	ABOUT
11.1	The Curve on 30 September 2026	3 min
11.2	A Straight Line Between Two Tenors	5 min
11.3	**`interpolate_yield`**, Docstring First	7 min
11.4	The Same With **`numpy.interp`**	3 min

	SECTION	ABOUT
11.5	The Curve the Holdings Were Built On	5 min
11.6	How Far the Straight Line Sits From the Smooth Curve	3 min
7	Bring It Home	1 min

Time. About 27 minutes for the lesson, then about 12 for the four exercises.

11.1 The Curve on 30 September 2026



EXCEL SIDE BY SIDE

JOB	PYTHON	EXCEL	RESULT
Chart the curve	`ax.plot(..., marker='o')`	select A1:B15, Insert, Scatter with Straight Lines and Markers	one series, `yield_pct`, tenors in years on the x axis
The same range as a Line chart	(none)	select A1:B15, Insert, Line	two series, the tenor column drawn as a line, 14 points evenly spaced

Fourteen points and none at 6.4 years: the yield there comes from a line drawn between two of them.

11.2 A Straight Line Between Two Tenors

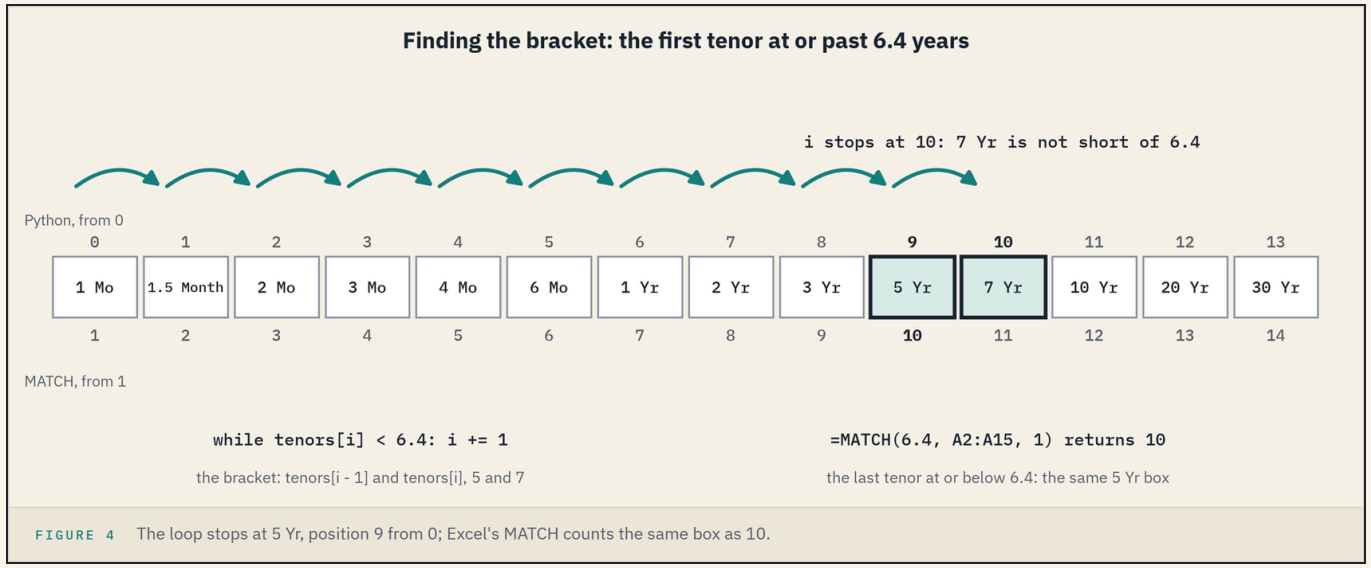


FIGURE 4 The loop stops at 5 Yr, position 9 from 0; Excel's MATCH counts the same box as 10.

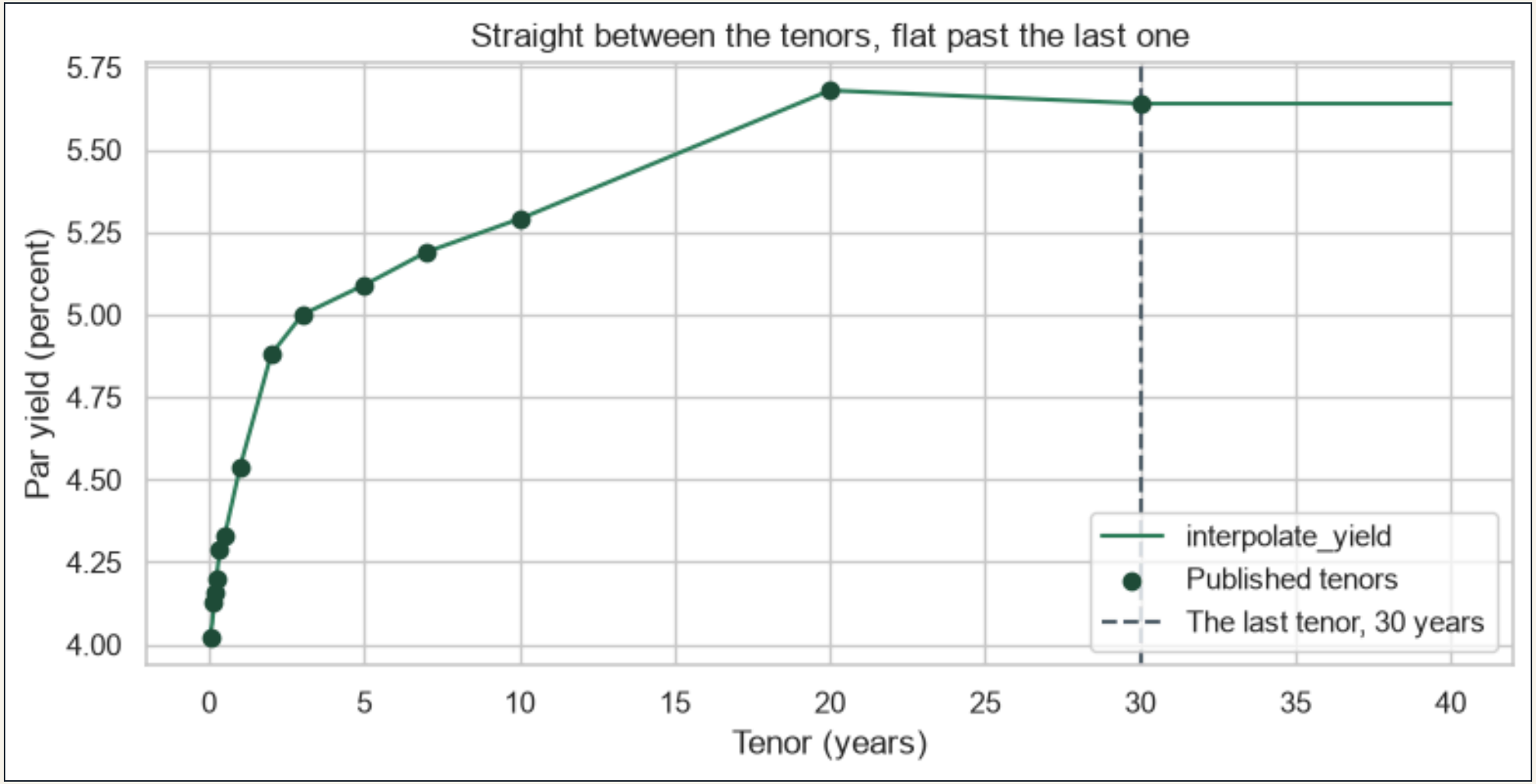
FIND THE BRACKET

```
i = 0
while tenors[i] < 6.4:
    i += 1
```

JOB	EXCEL
The bracket's position	`=MATCH(D1,A2:A15,1)`
The bracket's tenors	`=INDEX(A2:A15,MATCH(D1,A2:A15,1))`, and with `+1` after the `MATCH`

5.16 percent by hand, the same line as Excel's `FORECAST.LINEAR`, and a loop that finds the bracket for any maturity.

11.3 interpolate_yield, Docstring First



Flat Beyond the Ends, a Refusal for a Jumbled Curve

BEYOND THE ENDS

```
for tenor in [10, 0.02, 40]:
    result = bondmath.interpolate_yield(tenor, real_tenors, real_yields)
    print(f'{tenor:>5} years: {result:.2f} percent')
```

```
10 years: 5.29 percent
 0.02 years: 4.02 percent
   40 years: 5.64 percent
```

A JUMBLED CURVE

```
bondmath.interpolate_yield(4, [1, 3, 2], [3.7, 3.9, 3.8])
```

```
ValueError: curve tenors must ascend: 3 then 2
```

Excel's answers inside the curve, flat outside it, a refusal for a jumbled curve: 32 tests pass.

11.4 The Same With numpy.interp

EIGHT TENORS, TWO ANSWERS

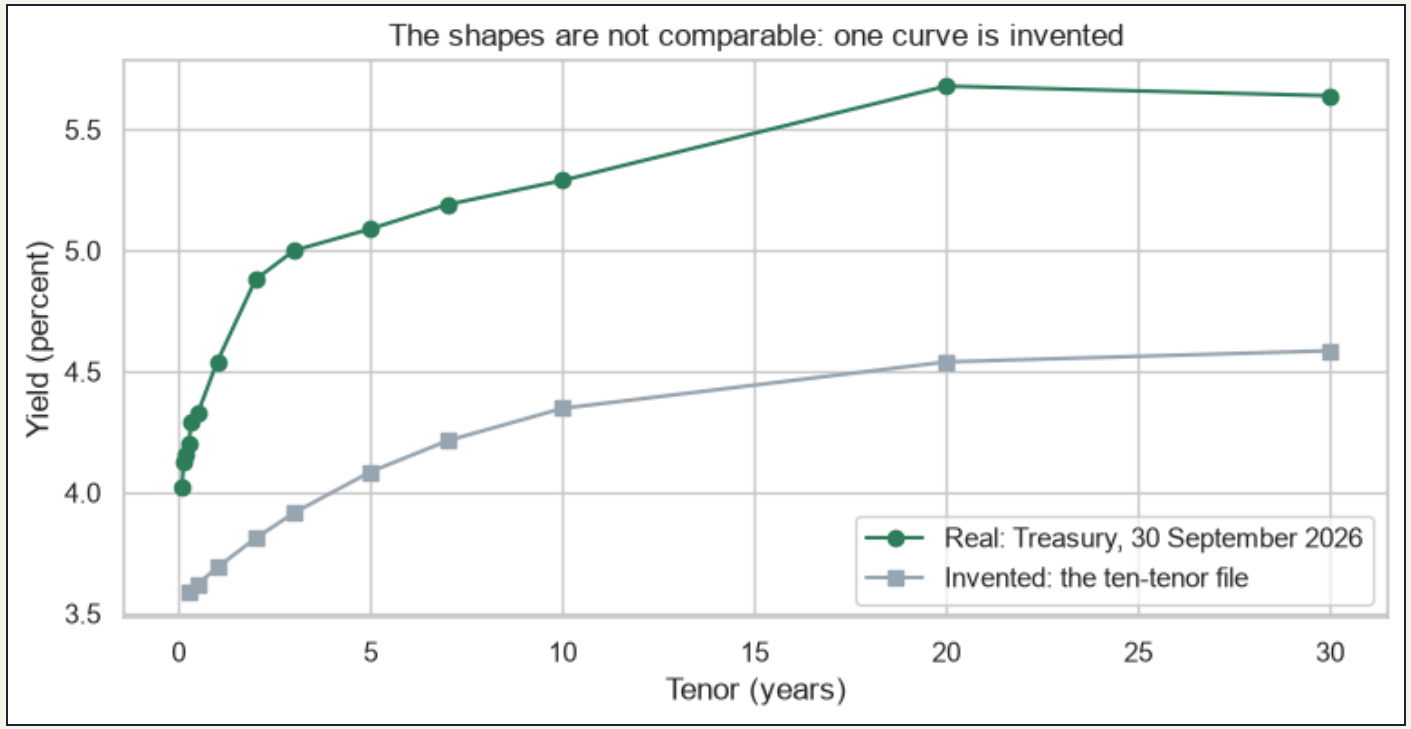
	TENOR_YEARS	OURS	NUMPY	DIFFERENCE
0	0.02	4.020	4.020	0.0
1	0.50	4.330	4.330	0.0
2	2.50	4.940	4.940	0.0
3	6.40	5.160	5.160	0.0
4	8.50	5.240	5.240	0.0
5	15.00	5.485	5.485	0.0
6	25.00	5.660	5.660	0.0
7	40.00	5.640	5.640	0.0

THE CODE

```
eight_tenors = [0.02, 0.5, 2.5, 6.4, 8.5, 15, 25, 40]
comparison = pd.DataFrame({'tenor_years': eight_tenors})
comparison['ours'] = [bondmath.interpolate_yield(t, real_tenors, real_yields)
                      for t in eight_tenors]
comparison['numpy'] = np.interp(eight_tenors, real_tenors, real_yields)
comparison['difference'] = comparison['ours'] - comparison['numpy']
comparison
```

Every tenor agrees, both flat ends included.

11.5 The Curve the Holdings Were Built On



THE TEN-TENOR FILE

	TENOR_YEARS	YIELD_PCT
0	0.25	3.587
1	0.50	3.622
2	1.00	3.690
3	2.00	3.811
4	3.00	3.916
5	5.00	4.086
6	7.00	4.214
7	10.00	4.348
8	20.00	4.540
9	30.00	4.586

The shapes are not comparable, because one curve is invented: it was written to give synthetic bonds believable yields, never fitted to Treasury's figures.

11.6 How Far the Straight Line Sits From the Smooth Curve



EVERY 0.01 YEARS

```
grid = np.linspace(0.25, 30, 2976)
gap_bp = (from_formula - from_lines) * 100
```

Bring It Home

COPY THE TWO FILES

```
Set-Location "$HOME\projects\my_bondmath"
Copy-Item "$env:TEMP\pfi_course2_11_k4w7q2xr\bondmath.py", "$env:TEMP\pfi_course2_11_k4w7q2xr\test_bondmath.py" .
python -m pytest -q
git diff --stat
git commit -am "Add interpolate_yield: linear between tenors, flat beyond the ends"
```

COMMIT THEM

```
!git -C "{work}" add bondmath.py test_bondmath.py
subject = 'Add interpolate_yield: linear between tenors, flat beyond the ends'
body = 'Week 3 reads a curve yield at each maturity; the tests hold it to numpy.interp.'
!git -C "{work}" commit -q -m "{subject}" -m "{body}"
!git -C "{work}" log --oneline -1
```

```
fa6939c Add interpolate_yield: linear between tenors, flat beyond the ends
```

You Can Now

- Turn Treasury's tenor labels into years and chart a curve at true distances.
- Read a curve between two tenors by hand, as a weighted average of the neighbours.
- Find the bracket with a loop, and with Excel's **`MATCH`** and **`INDEX`**.
- Read any tenor with **`interpolate_yield`**, tied to Excel's **`FORECAST.LINEAR`** inside the curve, flat beyond it, refusing a jumbled curve.
- Say what **`np.interp`** leaves to the caller.
- Put a number on what a straight line between tenors misses.
- See what Git leaves out with **`git status --short --ignored`**, and commit the module and its tests without the data.

NEXT
WEEK 3 · DAY 2

Spread Over the Curve

The holdings file's oas column rebuilt from the curve, bond by bond, and how the reading between tenors moves a spread.